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Dwyer Instruments - Primer On How Our Products Work

5-6 minutes

FACTORS AFFECTING MANOMETER PERFORMANCE AND USAGE

While the basic manometer principle of hydrostatic balance is inherently 100% accurate there are factors that can affect the actual pressure measurement obtained. Careful design and construction plus careful usage can eliminate or greatly reduce the effect of these factors.

CHARACTERISTICS OF INDICATING FLUID

No manometer can be read more accurately than the accuracy with which the specific gravity of the fluid is known. The fluid must also have good "wetting" characteristics and be capable of forming a consistent, well shaped meniscus in the indicating tube to facilitate accurate, repeatable readings.

The fluid used also affects the operating range of the manometer. Mercury being 13.6 times the weight of water will move 1/13.6th the distance water will move in response to a given pressure. Dwyer .826 sp. gr. gage oil being lighter than water will move about 1.2 times farther than water in response to a given pressure. This, obviously, expands the scale for easier, more precise reading.

Dwyer U-tube and well-type manometers are furnished with inch

scales for use with water or mercury or adjusted scales for use with .826 sp. gr. gage oil. Dwyer solid plastic vertical, inclined and inclined-vertical gages use gage oil.

Dwyer colored gage oil is a stable petroleum base oil with carefully controlled specific gravity which gives an excellent, consistent high visibility meniscus. Dwyer manometers for use with water are furnished with a fluorescein green concentrate which when added to water serves as a setting agent and a dye to improve the consistency and visibility of the meniscus for easier more accurate readability.

READABILITY

As we have seen, inclining the indicating tube and scale of a manometer, the use of lower specific gravity indicating fluids and the use of fluids that give a uniform, well defined meniscus facilitate accurate reading. Scales must be clear, sharp, accurate and easy to read. For accuracy, it is essential that the readings be made with the line of sight perpendicular to the fluid column to eliminate parallax error.



Fig. 2-5. Portion of the scale of a Dwyer No. 250.5 solid plastic inclined manometer shown full size. Parallax free reading is made by aligning meniscus with its reflection in the polished scale.

Dwyer solid plastic manometers assure parallax-free readings by the use of silk-screened scales on polished aluminum which reflect the image of the meniscus. When the meniscus and its reflection

are aligned, the line of sight is perpendicular to the fluid column at the meniscus and an accurate reading is assured. Smoothly machined bores further enhance the visibility of the meniscus.

LEVELING

Accurate readings with inclined and inclined-vertical manometers require that the inclined portion of the scale be at the exact angle for which it is designed. All Dwyer solid plastic inclined and inclined-vertical manometers are equipped with the integral, sensitive spirit levels to facilitate this requirement; most also have screw type leveling adjustment.

FACTORS IN DESIGN AND MANUFACTURE THAT AFFECT INCLINED MANOMETER ACCURACY

1. "Well Drop" (ratio of the area of the reservoir to the area of the indicating tube). As the fluid rises or falls in the indicating tube the level in the reservoir will fall or rise correspondingly and the scale must be compensated accordingly. Inaccuracies in the diameter of the reservoir or the indicating tube will create errors in this compensation. In glass tube instruments this error can only be minimized by a large ratio of reservoir to indicating tube area, by the use of precision bore tubing or by both. In the Dwyer solid plastic design, the use of machining techniques accurate to .0002" for wells and indicating bores reduces this error to insignificance.

2. Indicating Bore Straightness. This is a very real problem with glass tubing. Even with precision bore glass tubing, concentricity between I.D. and O.D. is difficult to control. Additionally, there is the problem of supporting the tube by its O.D. in perfect linearity with its I.D. Magnitude of error from this cause is, therefore, a function of the quality of this particular piece of precision bore tubing and the manner of tube mounting. Variations in accuracy from gage to gage

and in a given gage over a period of time can be anticipated as the tube is bumped or bent in use. In contrast, indicating tube bores up to 24" long in Dwyer solid plastic manometers are straight to within .002" over their entire length. It should also be noted that in the Dwyer design with a massive solid block of acrylic plastic, this straightness tolerance is effective for the life of the instrument.